

Week 4 · Laboratory Problems — build the guidance layer in Simulink

- Course: Sensor Signal Processing and Fusion · Department of Autonomous Vehicle System Engineering
- Time: **one hour**, immediately after the Week 4 lecture hour
- Three problems, in order.

What this hour is for

Weeks 1 to 3 were **told** what heading to hold. This hour builds the layer that decides it.

A complete Week 3 vessel is provided — heading autopilot, control allocation, hull — with one input left dangling: the commanded heading. That split is the point of the week. **Nothing inside the autopilot changes between Week 3 and Week 4**, and this model makes that seam visible.

Before starting

```
cd lectures/W04_simulink/problems
W04_P1_start           % creates W04_P1.slx — a Week 3 vessel, no guidance
W04_check(1)           % run this whenever, as often as needed
```

The solver is already set to fixed-step `ode4` at `h = 0.02` s. **Do not change it.**

★ Two requirements on every model

- `xlog` — To Workspace, `Structure With Time`, the plant's twelve states
- `glog` — To Workspace, `Structure With Time`, carrying $[\pi_p; y_e^p; \psi_d]$ in **radians and metres, in that order**

The guidance log is required because Problems 1 and 3 are about quantities that never appear in the state vector. A model can put the vessel in the right place with a wrong y_e^p , and only `glog` can tell.

The waypoints are already in the workspace as `WP_N` and `WP_E`. **Leg 1** runs from $\mathbf{p}_1^n = (0, 0)$ to $\mathbf{p}_2^n = (60, 0)$ — due north, so π_p is exactly zero. That is deliberate: a leg whose angle is zero makes a sign error in the rotation visible immediately.

Problem 1 · The LOS law (25 minutes)

Build. A guidance block taking the position and producing ψ_d .

$$\begin{aligned}\pi_p &= \text{atan2}(y_{i+1}^n - y_i^n, x_{i+1}^n - x_i^n) \\ y_e^p &= -(x^n - x_i^n) \sin \pi_p + (y^n - y_i^n) \cos \pi_p \\ \psi_d &= \pi_p - \tan^{-1}\left(\frac{y_e^p}{\Delta}\right)\end{aligned}$$

Leg 1 only; no waypoint switching this hour.

⚠ Two different arctangents, and they are not interchangeable

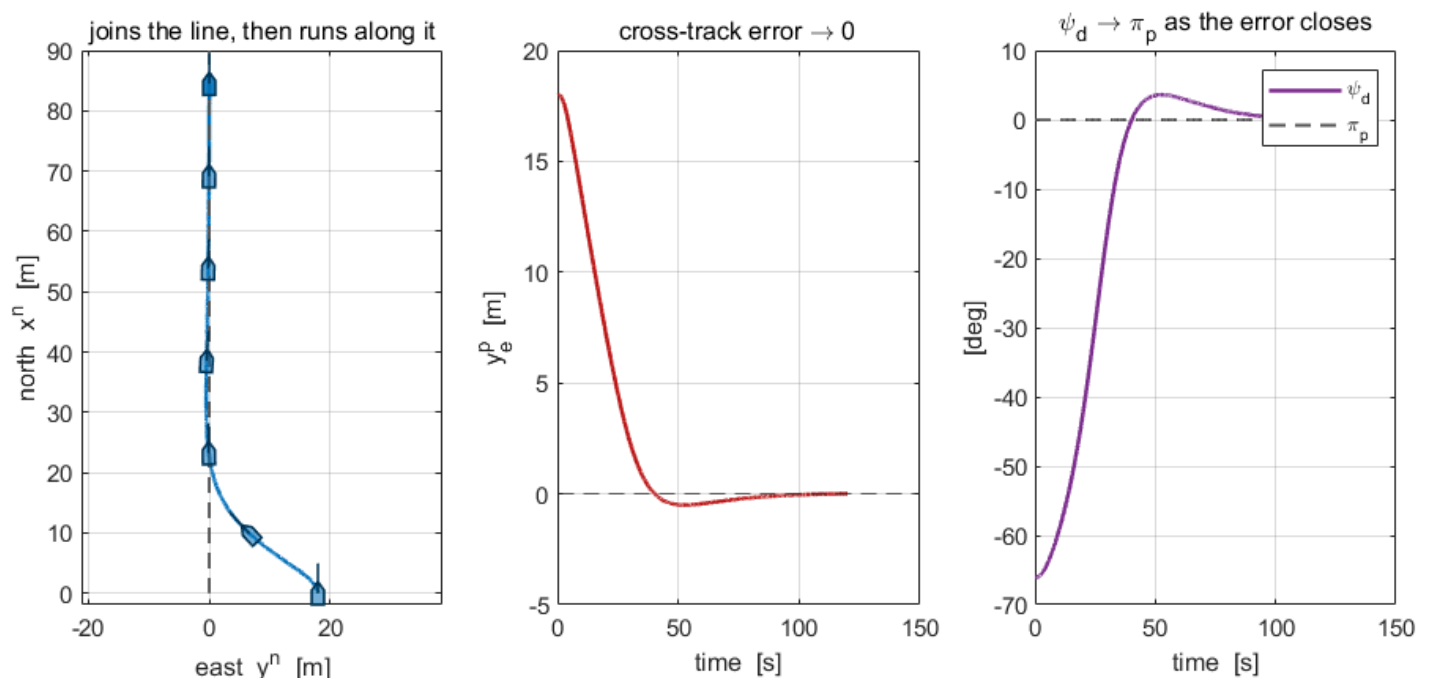
π_p **must** use the two-argument `atan2` — a leg can point into any of the four quadrants, and the one-argument form folds two of them onto the others.

The correction term may use the one-argument `atan`, because $\Delta > 0$ always and the aim-point vector therefore never leaves the right half-plane of $\{p\}$.

Verify. `w04_check(1)`. The vessel starts **18 m** east of the leg.

What the checker expects	
π_p on leg 1	exactly 0 rad
initial y_e^p	18 m
settled y_e^p	0
settled ψ_d	0 rad, i.e. $\psi_d \rightarrow \pi_p$

What a correct model produces



Reading the figure	
left	the vessel curves onto the dashed path and then runs along it. Hulls are drawn so the heading is visible
middle	y_e^p falls smoothly to zero without crossing — with $\Delta = 8$ m it does not overshoot
right	ψ_d starts well off π_p and converges onto it as the error closes
the check	if the initial y_e^p comes out -18 m, the two sine terms have swapped sign

Problem 2 · Aiming at a point is not following a path (20 minutes)

Build. Add a second law to the same block, selected by a flag:

$$\psi_d = \text{atan2}(y_{i+1}^n - y^n, x_{i+1}^n - x^n)$$

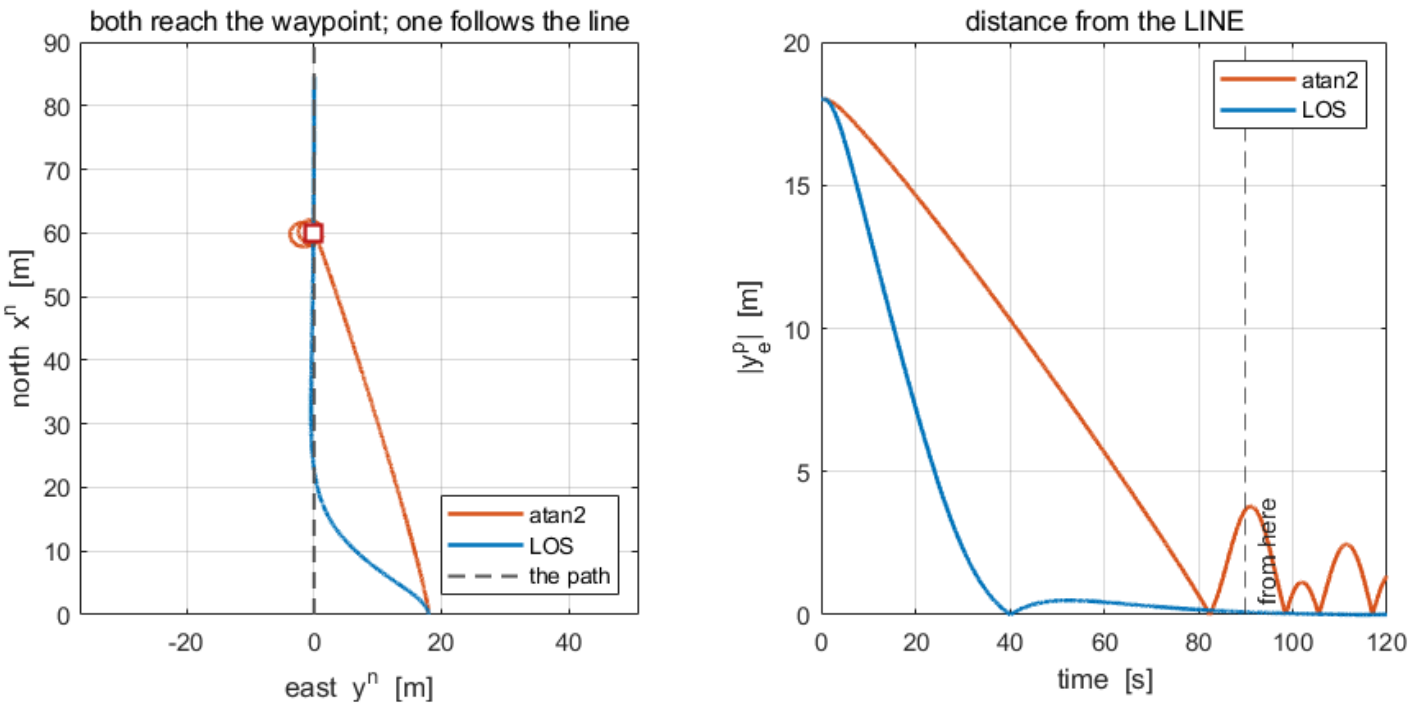
Both laws must share one plant, one autopilot and one allocation. If they do not, the comparison measures something other than the guidance.

Verify. `w04_check(2)` , same start.

	worst $ y_e^p $ after 90 s
LOS	0.09 m
atan2	3.78 m

Both still reach the waypoint. That is essential to the argument: `atan2` is **not broken**.

What a correct model produces



Reading the figure	
left	both tracks end at the red waypoint marker. Only the blue one lies on the dashed path while doing so
right	distance from the line . The orange trace stays an order of magnitude above the blue one
the check	if the two tracks are identical, the law flag is not reaching the guidance block

The point. `atan2` regulates the distance to a **point**, and a point carries no information about the line it sits on. No autopilot gain fixes this, because the path does not appear anywhere in the law. **It is answering a different question.**

Problem 3 · A current, and the offset that stays (15 minutes)

Build. Nothing. Set $V_c = 0.3$ m/s and $\beta_c = 90^\circ$ and run the LOS law again.

Predict before running. Section 4-7 derives

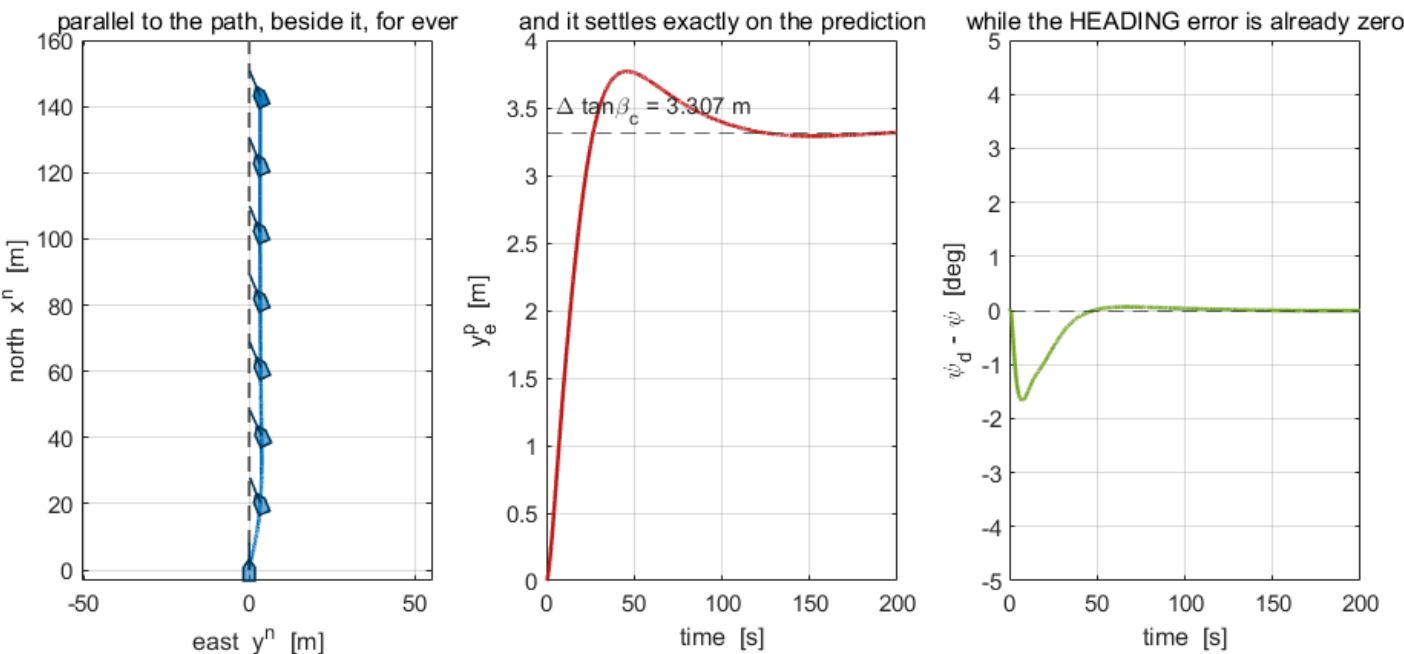
$$y_e^{p,ss} = \Delta \tan \beta_c$$

Measure β_c from the run as `atan2(v,u)` and work out what the offset should be.

Verify. `w04_check(3)` .

What the checker expects	
settled y_e^p	equal to $\Delta \tan \beta_c$ to within 0.15 m
heading error while that offset persists	0

What a correct model produces



Reading the figure	
left	the track runs parallel to the path and beside it. The hulls lean upstream
middle	y_e^p rises and settles exactly on the dashed prediction $\Delta \tan \beta_c$
right	the heading error is 0 the whole time the offset persists
the check	if the offset keeps growing, the guidance is reading a stale position; if it returns to zero, an integrator has crept in

The point. The loop is doing exactly what it was asked. The heading error is **already zero**, so there is nothing left for a larger autopilot gain to act on. **The law is not short of authority; it is short of terms.** Sections 4-8 and 4-9 are two ways of supplying the missing one, and neither of them uses gain.

Marking

	Weight	What is being marked
Problem 1	40	<code>w04_check(1)</code> passes; the reason π_p needs <code>atan2</code> and the correction does not is stated
Problem 2	30	<code>w04_check(2)</code> passes; the sentence "it regulates a point, not a line" is argued rather than quoted
Problem 3	30	<code>w04_check(3)</code> passes; the predicted $\Delta \tan \beta_c$ was written down before running

If something goes wrong

Symptom	Cause	Fix
The model has no To Workspace block whose variable name is <code>glog</code>	only <code>xlog</code> was added	add the second log, three signals, <code>Structure With Time</code>
<code>glog</code> has 1 column; 3 were expected	a matrix signal reached the log	the MATLAB Function output is 3×1 ; insert a Reshape set to <code>1-D array</code>
Initial y_e^p is -18 m	the two sine terms are swapped	$y_e^p = -\Delta x \sin \pi_p + \Delta y \cos \pi_p$
The vessel turns away from the path	the sign of the correction is reversed	the law subtracts the arctangent from π_p
The vessel spirals	π_p was computed with one-argument <code>atan</code>	use <code>atan2</code>
ψ_d jumps by 360°	the command was wrapped in the guidance	do not wrap it there; the autopilot wraps the error , which is the only place a wrap belongs

Reference answers are in `../solutions/`. Read them **after** attempting the problem.